ID: beca03de

A rectangle has a length that is 15 times its width. The function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

A. The length of the rectangle, in feet

B. The area of the rectangle, in square feet

C. The difference between the length and the width of the rectangle, in feet

D. The width of the rectangle, in feet

ID: 4e18fc5d

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v, w, and x. Which equation correctly expresses w in terms of v and x?

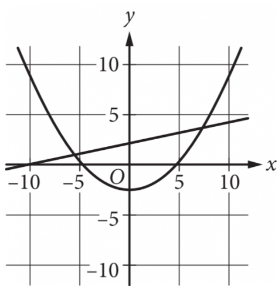
A. $w = -150vx$

B. $w = -\frac{150v}{x}$

C. $w = -\frac{x}{150v}$

D. $w = v + 150x$

ID: a5663025



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the *xy*-plane above. How many solutions does this system have?

A. 0

B. 1

C. 2

D. 3

ID: d0a7871e

$$y = x + 1$$
$$y = x^2 + x$$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x ?

A. -1

B. 0

C. 2

D. 3

ID: b8caaf84

If $p=3x+4$ and $v=x+5$, which of the following is equivalent to $pv-2p+v$?

A. $3x^2+12x+7$

B. $3x^2+14x+17$

C. $3x^2+19x+20$

D. $3x^2+26x+33$

ID: dd4ab4c4

$$4a^2+20ab+25b^2$$

Which of the following is a factor of the polynomial above?

A. $a+b$

B. $2a+5b$

C. $4a+5b$

D. $4a+25b$

ID: 7f81d0c3

$$x^2 - x - 1 = 0$$

What values satisfy the equation above?

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$

ID: 52931bfa

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x>7$?

A. $\frac{x-7}{5}$

B. $\frac{8x-3}{2}$

C. $\frac{8x^2-3x-14}{2x-14}$

D. $\frac{8x^2-3x-77}{2x-14}$

ID: 39714777

$$p(x) + 57 = x^2$$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p. What is the minimum value of the function p?

A. $-3{,}249$

B. -57

C. 57

D. $3{,}249$

ID: cb29c54c

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y, in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the *xy*-plane?

A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.

B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.

D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

ID: ad2ec615

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

A. $(x^2+1)(x^2-6)$

B. $(x^2+2)(x^2-3)$

C. $(x^2+3)(x^2-2)$

D. $(x^2+6)(x^2-1)$

ID: 42c71eb5

$$(2x+5)^2-(x-2)+2(x+3)$$

Which of the following is equivalent to the expression above?

A. $4x^2+21x+33$

B. $4x^2+21x+29$

C. $4x^2+x+29$

D. $4x^2+x+33$

ID: 52b1700c

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Rosa opened the account and the total amount n, in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A. $n=(1+604)^t$

B. $n=(1+0.004)^t$

C. $n=604(1+0.004)^t$

D. $n=0.004(1+604)^t$

ID: b4acba95

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly two

B. Exactly one

C. Zero

D. Infinitely many

ID: ff8c5844

x	$g(x)$
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g, the table shows four values of x and their corresponding values of $g(x)$. Which equation defines g?

A. $g(x) = -25^x$

B. $g(x) = -\left(\frac{1}{25}\right)^x$

C. $g(x) = 25^x$

D. $g(x) = \left(\frac{1}{25}\right)^x$

ID: a05bd3a4

Which of the following expressions is equivalent to $x^2 - 5$?

A. $(x+\sqrt{5})^2$

B. $(x-\sqrt{5})^2$

C. $(x+\sqrt{5})(x-\sqrt{5})$

D. $(x+5)(x-1)$

ID: f423771c

x	$h(x)$
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A. $h(x) = 1.12(0.23)^x$

B. $h(x) = 1.12(1.23)^x$

C. $h(x) = 1.23(0.12)^x$

D. $h(x) = 1.23(1.12)^x$

ID: 1d3c5c95

$$f(x) = 4{,}000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \le 7$. Which of the following is the best interpretation of $4{,}000$ in this context?

A. The estimated initial beetle population for this species and area in 2012

B. The estimated beetle population for this species and area 7 years after 2012

C. The estimated percent decrease in the beetle population for this species and area each year after 2012

D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

ID: ae05d37b

The function $f(t) = 40{,}000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

A. 2

B. 790

C. $1{,}580$

D. $40{,}000$

ID: 02add2d2

A company has a newsletter. In January 2018, there were $1{,}300$ customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c, subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

A. $c = 1{,}300(0.07)^m$

B. $c = 1{,}300(1.07)^m$

C. $c = 1{,}300(1.7)^m$

D. $c = 1{,}300(7)^m$

ID: 3206b905

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

A. $x\left(7x^{10} - 7x^9 + 87x\right)$

B. $x\left(8^{10} - 8^9 + 88\right)$

C. $8x\left(x^{10} - x^9 + 11x\right)$

D. $8x\left(x^9 - x^8 + 11\right)$

ID: f89af023

A rectangular volleyball court has an area of 162 square meters. If the length of the court is twice the width, what is the width of the court, in meters?

A. 9

B. 18

C. 27

D. 54

ID: 768b60d2

For the exponential function f, the value of $f(0)$ is c, where c is a constant. Of the following equations that define the function f, which equation shows the value of c as the coefficient or the base?

A. $f(x) = 22(1.5)^{x+1}$

B. $f(x) = 33(1.5)^{x}$

C. $f(x) = 49.5(1.5)^{x-1}$

D. $f(x) = 74.25(1.5)^{x-2}$

ID: e53add44

$$S(n) = 38{,}000a^n$$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

A. 0.04

B. 0.4

C. 1.04

D. 1.4

ID: b4a6ed81

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r?

ID: 4618501a

$$f(x) = 3{,}000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of $3{,}000$ in this context?

A. The estimated percent decrease in the population for this species and area every 8 years after 2008

B. The estimated percent decrease in the population for this species and area each year after 2008

C. The estimated population for this species and area 8 years after 2008

D. The estimated initial population for this species and area in 2008

ID: cc776a04

Which of the following is an equivalent form of $(1.5x-2.4)^2-(5.2x^2-6.4)$?

A. $-2.2x^2+1.6$

B. $-2.2x^2+11.2$

C. $-2.95x^2-7.2x+12.16$

D. $-2.95x^2-7.2x+0.64$

ID: ff2c1431

$$7m = 5(n + p)$$

The given equation relates the positive numbers m, n, and p. Which equation correctly gives n in terms of m and p?

A. $n = \frac{5p}{7m}$

B. $n = \frac{7m}{5} - p$

C. $n = 5(7m) + p$

D. $n = 7m - 5 - p$

ID: 926c246b

$$D = 5{,}640(1.9)^t$$

The equation above estimates the global data traffic D, in terabytes, for the year that is t years after 2010. What is the best interpretation of the number 5,640 in this context?

A. The estimated amount of increase of data traffic, in terabytes, each year

B. The estimated percent increase in the data traffic, in terabytes, each year

C. The estimated data traffic, in terabytes, for the year that is t years after 2010

D. The estimated data traffic, in terabytes, in 2010

ID: 6bdcac03

$$x^2 = -841$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

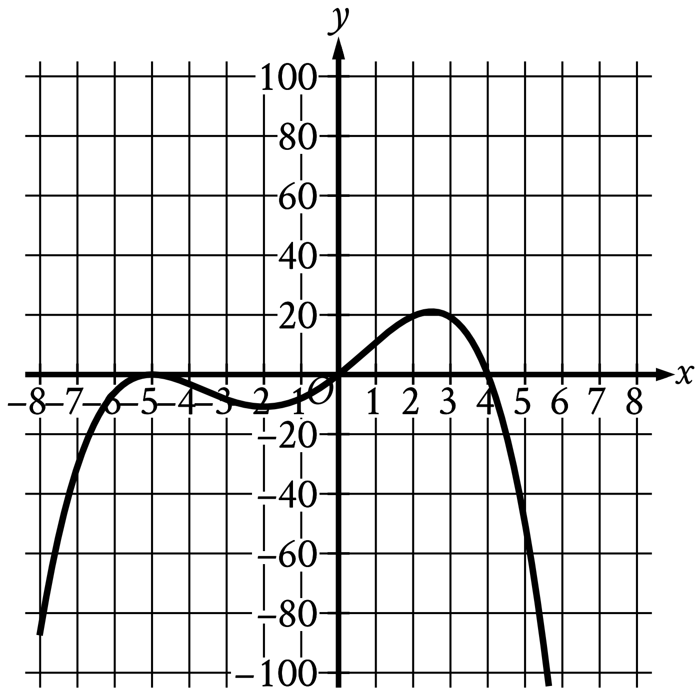
D. Zero

ID: 3d7d7534

$$(d-30)(d+30)-7=-7$$

What is a solution to the given equation?

ID: 252a3b3a



Which of the following could be the equation of the graph shown in the *xy*-plane?

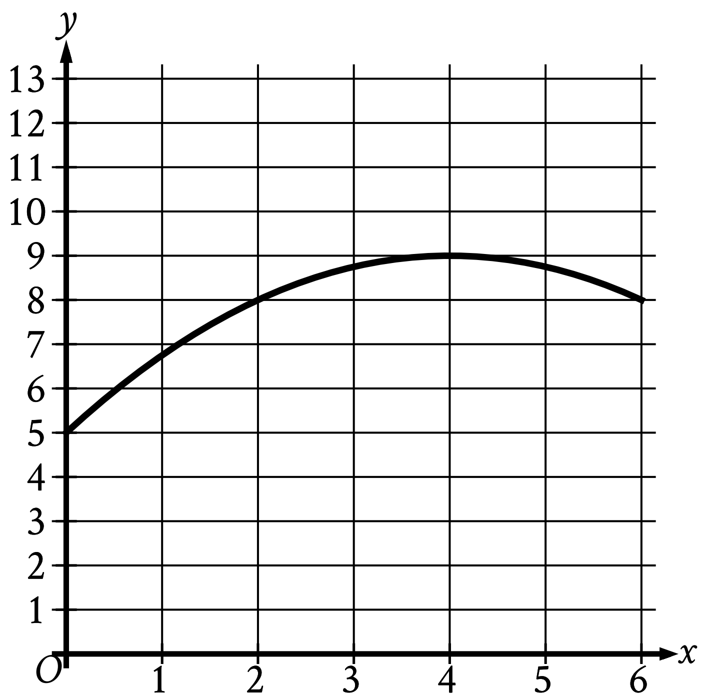
A. $y = -\frac{1}{10}x(x-4)(x+5)$

B. $y = -\frac{1}{10}x(x-4)(x+5)^2$

C. $y = -\frac{1}{10}x(x-5)(x+4)$

D. $y = -\frac{1}{10}x(x-5)^2(x+4)$

ID: 95d1c344



The graph models the number of active projects a company was working on x months after the end of November 2012, where $0 \leq x \leq 6$. According to the model, what is the predicted number of active projects the company was working on at the end of November 2012?

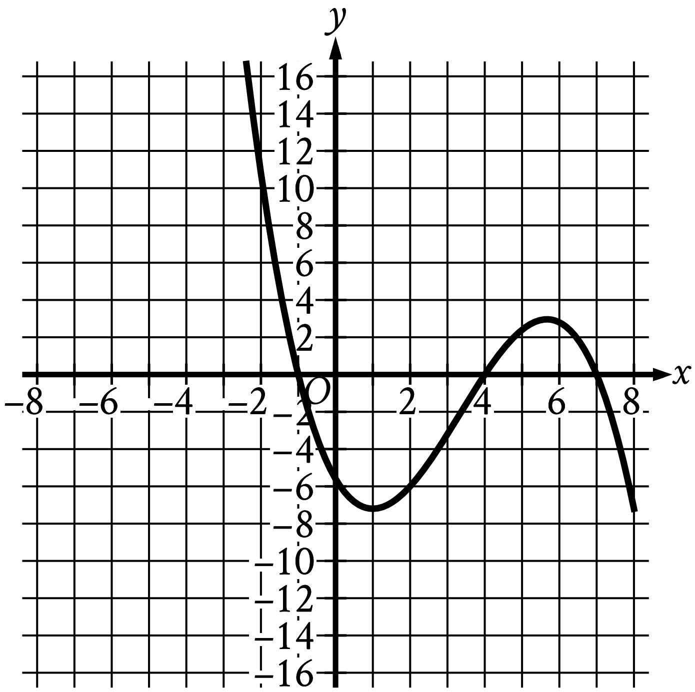
A. 0

B. 5

C. 8

D. 9

ID: cc6ccd71



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x does $f(x) = 0$?

A. One

B. Two

C. Three

D. Four

ID: 911383f2

$$(x-4)(x+2)(x-1)=0$$

What is the product of the solutions to the given equation?

A. 8

B. 3

C. -3

D. -8

ID: b80d10d7

$$\frac{2(x+1)}{x+5} = 1 - \frac{1}{x+5}$$

What is the solution to the equation above?

A. 0

B. 2

C. 3

D. 5

ID: fde6f3bb

$$g(x) = \frac{3}{5}x + \frac{7}{6}$$
$$h(x) = 6x - 5$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

A. $\frac{18x^2}{5} - \frac{35}{6}$

B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$

C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$

D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

ID: d4950429

A rectangle has a length of x units and a width of $(x - 15)$ units. If the rectangle has an area of 76 square units, what is the value of x?

A. 4

B. 19

C. 23

D. 76

ID: 752055d1

A scientist initially measures $12{,}000$ bacteria in a growth medium. 4 hours later, the scientist measures $24{,}000$ bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r?

A. $\frac{1}{12{,}000}$

B. $\frac{1}{4}$

C. 4

D. $12{,}000$

ID: 3148fe3e

$$x^2 + y + 10 = 10$$
$$8x + 16 - y = 0$$

The solution to the given system of equations is (x, y). What is the value of x?

A. -16

B. -4

C. 2

D. 8

ID: fcdf87b7

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

ID: a520ba07

$$\sqrt[3]{x^3y^6}$$

Which of the following expressions is equivalent to the expression above?

A. y^2

B. xy^2

C. y^3

D. xy^3

ID: 5b6af6b1

Which expression is equivalent to $(d-6)\left(8d^2-3\right)$?

A. $8d^3-14d^2-3d+18$

B. $8d^3-17d^2+48$

C. $8d^3-48d^2-3d+18$

D. $8d^3-51d^2+48$

ID: 652054da

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed *s*, in knots, of an ocean wave, where *p* represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

A. $p = \frac{2}{3}s$

B. $p = \frac{3}{2}s$

C. $p = \frac{2}{3} + s$

D. $p = \frac{3}{2} + s$

ID: 0380bbdc

If $4\sqrt{2x} = 16$, what is the value of $6x$?

A. 24

B. 48

C. 72

D. 96

ID: a255ae72

If $x^2=a+b$ and $y^2=a+c$, which of the following is equal to $(x^2-y^2)^2$?

A. $a^2-2ac+c^2$

B. $b^2-2bc+c^2$

C. $4a^2-4abc+c^2$

D. $4a^2-2abc+b^2c^2$

ID: dd3b1e1a

$$f(x) = x^5 + 9x + 17$$

For the given function f, the graph of $y = f(x)$ in the *xy*-plane passes through the point $(0, b)$, where b is a constant. What is the value of b?

ID: 95ed0b69

$$p=\frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p, k, and j. Which equation correctly expresses $4j+9$ in terms of p and k?

A. $4j+9=\frac{k}{p}$

B. $4j+9=kp$

C. $4j+9=k-p$

D. $4j+9=\frac{p}{k}$

ID: 463eec13

If $x \neq 0$, which of the following expressions is equivalent to $\dfrac{\sqrt{16x^4y^8}}{x^3}$?

A. $8x^2y^4$

B. $4xy^4$

C. $4x^{-2}y^2$

D. $4x^{-1}y^4$

ID: 341ba5db

$$g(x) = x^2 + 55$$

What is the minimum value of the given function?

A. 0

B. 55

C. 110

D. $3{,}025$

ID: 6e02cd78

In the *xy*-plane, what is the *y*-coordinate of the point of intersection of the graphs of $y=(x-1)^2$ and $y=2x-3$?

ID: 15c364bf

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(87) = 23$ in this context?

A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.

B. The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.

C. The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.

D. Approximately 87 grams of the sample remains 23 years after the sample starts to decay.

ID: f28944ff

$$q(x) = 32(2^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q?

A.

x	-1	0	1
$q(x)$	-64	0	64

B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64

C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64

D.

x	-1	0	1
$q(x)$	16	32	64

ID: 11ccf3e1

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m?

A. $k = \frac{m - 14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j - m}{5}$

D. $k = 5m - 14j$

ID: 50e40f08

$$f(x) = (x + 6)(x - 4)$$

If the given function f is graphed in the *xy*-plane, where $y = f(x)$, what is the *x*-coordinate of an *x*-intercept of the graph?

ID: c4259674

The function f is defined by $f(x) = 4x^{-1}$. What is the value of $f(21)$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$

ID: be0c419e

Immanuel purchased a certain rare coin on January 1. The function $f(x)=65(1.03)^x$, where $0\leq x\leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement "$f(8)$ is approximately equal to 82" in this context?

A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.

B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.

C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.

D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

ID: 13e57f0a

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

A. $\frac{7}{4}$

B. $\frac{9}{4}$

C. 4

D. 7

ID: a1bf1c4e

$$x^2 + 6x + 4$$

Which of the following is equivalent to the expression above?

A. $(x + 3)^2 + 5$

B. $(x + 3)^2 - 5$

C. $(x - 3)^2 + 5$

D. $(x - 3)^2 - 5$

ID: 802549ac

$$(x+2)(x+3)=(x-2)(x-3)+10$$

Which of the following is a solution to the given equation?

A. 1

B. 0

C. -2

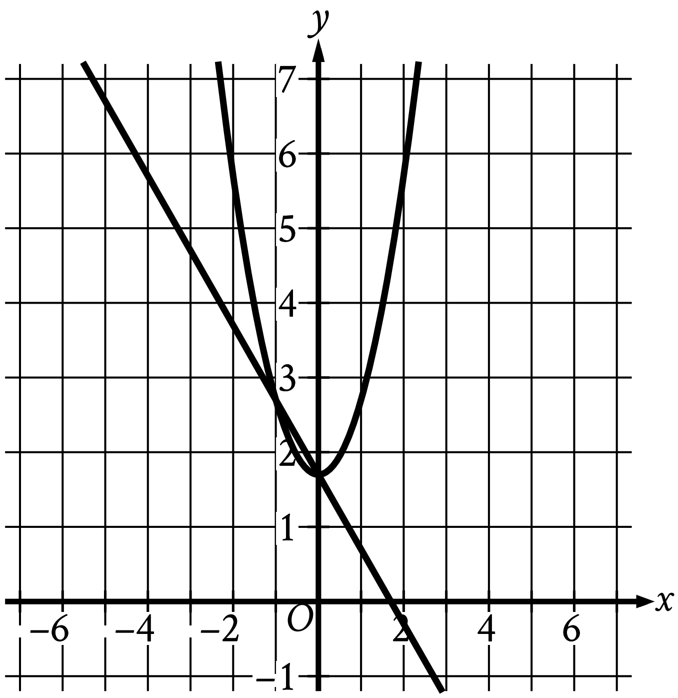
D. -5

ID: 75a32330

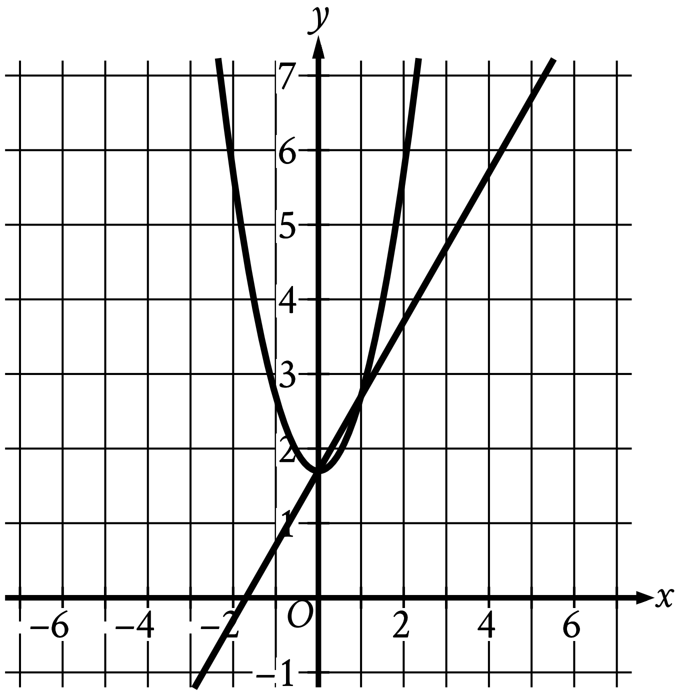
$$y = x^2 + 1.7$$
$$y = 1.7 - x$$

Which graph represents the given system of equations?

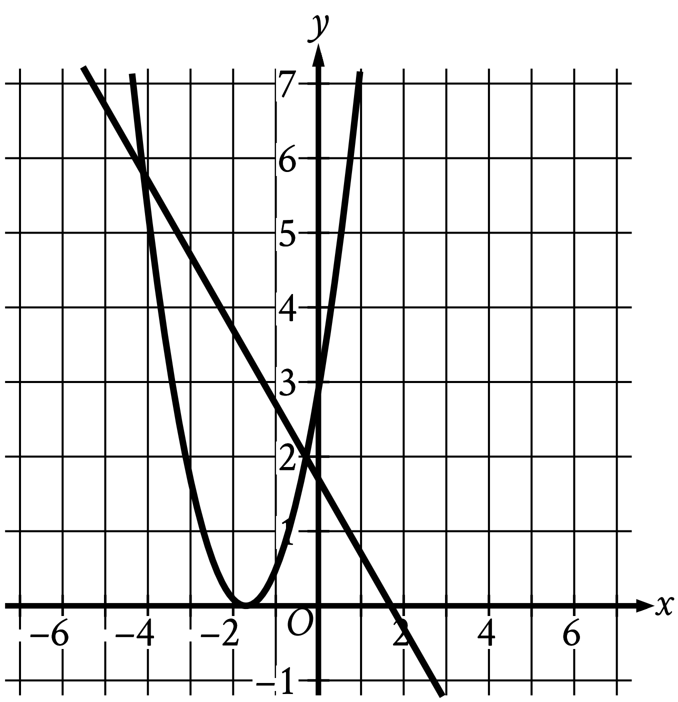
A.



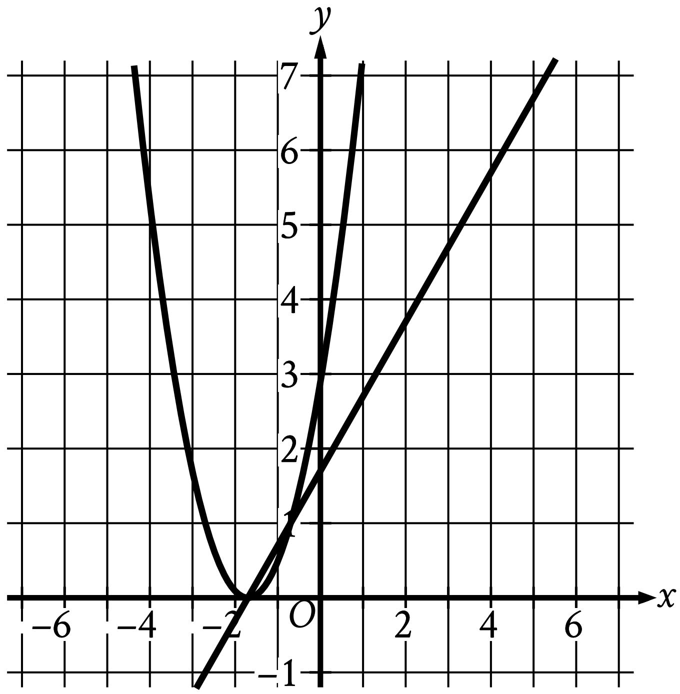
B.



C.



D.



ID: e1f9000d

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is $3{,}364$ feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

A. 7.25

B. 14.50

C. 105.13

D. 210.25

ID: a4f61d75

$$x^2 - ax + 12 = 0$$

In the equation above, *a* is a constant and $a > 0$. If the equation has two integer solutions, what is a possible value of *a* ?

ID: a31417d1

From 2005 through 2014, the number of music CDs sold in the United States declined each year by approximately 15% of the number sold the preceding year. In 2005, approximately 600 million CDs were sold in the United States. Of the following, which best models *C*, the number of millions of CDs sold in the United States, *t* years after 2005?

A. $C = 600(0.15)^t$

B. $C = 600(0.85)^t$

C. $C = 600(1.15)^t$

D. $C = 600(1.85)^t$

ID: 6d04c89d

The expression $\frac{24}{6x+42}$ is equivalent to $\frac{4}{x+b}$, where *b* is a constant and $x > 0$. What is the value of *b*?

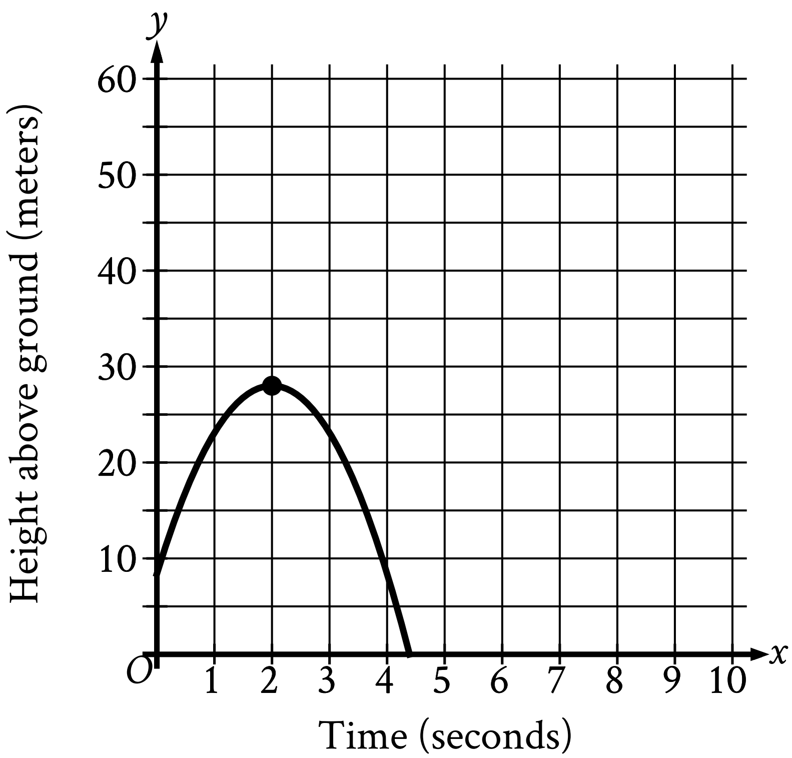
A. 7

B. 10

C. 24

D. 252

ID: 197bed38



An object was launched upward from a platform. The graph shown models the height above ground, y, in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

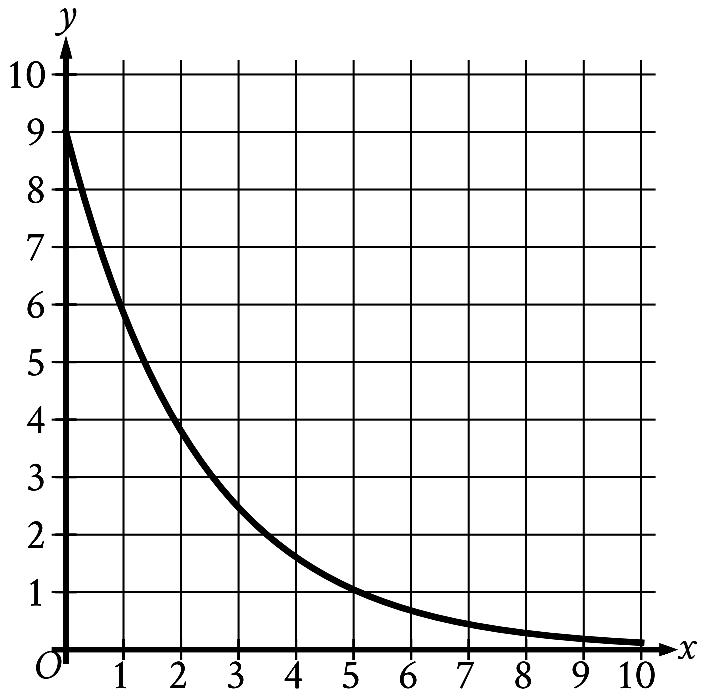
A. From $x = 0$ to $x = 2$

B. From $x = 0$ to $x = 4$

C. From $x = 2$ to $x = 3$

D. From $x = 3$ to $x = 4$

ID: db888cd6



The graph gives the estimated number of catalogs y, in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of 1992, where $0 \leq x \leq 10$. Which statement is the best interpretation of the *y*-intercept in this context?

A. The estimated total number of catalogs the company sent to its customers during the first 10 years was $9{,}000$.

B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.

D. The estimated number of catalogs the company sent to its customers at the end of 1992 was $9{,}000$.

ID: 6ecdbcb4

$$f(x) = (x+6)(x+5)(x-4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

ID: 203774bc

The product of two positive integers is 546. If the first integer is 11 greater than twice the second integer, what is the smaller of the two integers?

A. 7

B. 14

C. 39

D. 78

ID: c4cd5bcc

In the *xy*-plane, the *y*-coordinate of the *y*-intercept of the graph of the function f is c. Which of the following must be equal to c ?

A. $f(0)$

B. $f(1)$

C. $f(2)$

D. $f(3)$

ID: dc77e0dc

$$f(t) = 500(0.5)^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

A. 500

B. 12

C. 5

D. 2

ID: 062f86db

$$5x^2 - 37x - 24 = 0$$

What is the positive solution to the given equation?

A. $\frac{3}{5}$

B. 3

C. 8

D. 37

ID: 735a0a00

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y, in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $(x, y) = (1, 83)$ in this context?

A. The company's estimated stock price increased $\$83$ every day after the new product launched.

B. The company's estimated stock price increased $\$1$ every 83 days after the new product launched.

C. 1 day after the new product launched, the company's estimated stock price is $\$83$.

D. 83 days after the new product launched, the company's estimated stock price is $\$1$.

ID: 68607eca

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function f gives the predicted number of views x days after April 1. Which equation defines f?

A. $f(x) = 233(0.70)^{\frac{x}{2}}$

B. $f(x) = 233(0.70)^{2x}$

C. $f(x) = 233(1.70)^{\frac{x}{2}}$

D. $f(x) = 233(1.70)^{2x}$

ID: 717a1964

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

ID: 78d5f91a

$$f(x) = x^3 + 3x^2 - 6x - 1$$

For the function *f* defined above, what is the value of $f(-1)$?

A. -11

B. -7

C. 7

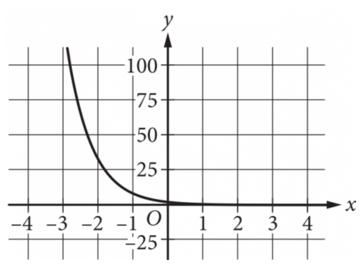
D. 11

ID: d675744f

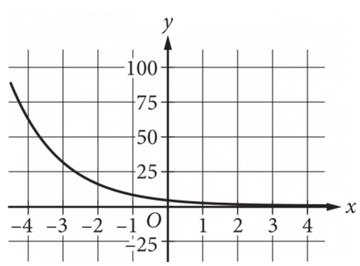
$y = 4(2^x)$

Which of the following is the graph in the *xy*-plane of the given equation?

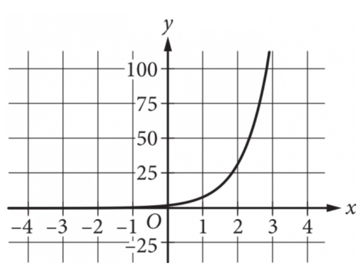
A.



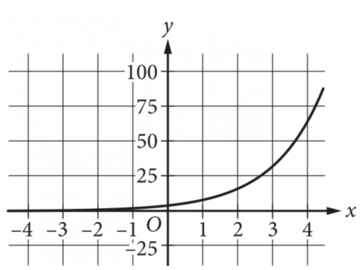
B.



C.



D.



ID: 67f4b449

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet $\left(\text{ft}^2\right)$, if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1{,}176$?

A. If the width of the rectangle is $14\ \text{ft}$, then the area of the rectangle is $1{,}176\ \text{ft}^2$.

B. If the width of the rectangle is $14\ \text{ft}$, then the length of the rectangle is $1{,}176\ \text{ft}$.

C. If the width of the rectangle is $1{,}176\ \text{ft}$, then the length of the rectangle is $14\ \text{ft}$.

D. If the width of the rectangle is $1{,}176\ \text{ft}$, then the area of the rectangle is $14\ \text{ft}^2$.

ID: f880f910

The area of a triangle is 270 square centimeters. The length of the base of the triangle is 12 centimeters greater than the height of the triangle. What is the height, in centimeters, of the triangle?

A. 15

B. 18

C. 30

D. 36

ID: fad2f98c

$$3x(x-4)(x+5)=0$$

What is one of the solutions to the given equation?

A. -4

B. 0

C. 3

D. 5

ID: 44076c7d

x	$f(x)$
-1	10
0	14
1	20

For the quadratic function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f?

A. $f(x) = 3x^2 + 3x + 14$

B. $f(x) = 5x^2 + x + 14$

C. $f(x) = 9x^2 - x + 14$

D. $f(x) = x^2 + 5x + 14$

ID: a267bd29

$$w^2 + 12w - 40 = 0$$

Which of the following is a solution to the given equation?

A. $6 - 2\sqrt{19}$

B. $2\sqrt{19}$

C. $\sqrt{19}$

D. $-6 + 2\sqrt{19}$

ID: f44a29a8

An object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. What is the speed, in meters per second, of an object with a mass of 4 kilograms and kinetic energy of 18 joules?

A. 3

B. 6

C. 9

D. 36

ID: d71f6dbf

The height, in feet, of an object x seconds after it is thrown straight up in the air can be modeled by the function $h(x) = -16x^2 + 20x + 5$. Based on the model, which of the following statements best interprets the equation $h(1.4) = 1.64$?

A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.

B. The height of the object 1.64 seconds after being thrown straight up in the air is 1.4 feet.

C. The height of the object 1.64 seconds after being thrown straight up in the air is approximately 1.4 times as great as its initial height.

D. The speed of the object 1.4 seconds after being thrown straight up in the air is approximately 1.64 feet per second.

ID: 630897df

The speed of sound in dry air, v, can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

A. $T = \frac{v + 0.606}{331.3}$

B. $T = \frac{v - 0.606}{331.3}$

C. $T = \frac{v + 331.3}{0.606}$

D. $T = \frac{v - 331.3}{0.606}$

ID: 6676f055

$$f(\theta) = -0.28(\theta - 27)^2 + 880$$

An engineer wanted to identify the best angle for a cooling fan in an engine in order to get the greatest airflow. The engineer discovered that the function above models the airflow $f(\theta)$, in cubic feet per minute, as a function of the angle of the fan θ, in degrees. According to the model, what angle, in degrees, gives the greatest airflow?

A. -0.28

B. 0.28

C. 27

D. 880

ID: 5805e747

Which expression is equivalent to $\left(7x^3+7x\right)-\left(6x^3-3x\right)$?

A. x^3+10x

B. $-13x^3+10x$

C. $-13x^3+4x$

D. x^3+4x

ID: 29ed5d39

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n, where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p?

A. $n = \frac{p-20}{16}$

B. $n = \frac{p}{16} + 20$

C. $n = \frac{p}{16} - 20$

D. $n = \frac{16}{p-20}$

ID: 90bcaa61

The function $f(t) = 60{,}000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

ID: 895628b5

$$y=(x-2)(x+4)$$
$$y=6x-12$$

Which ordered pair (x, y) is the solution to the given system of equations?

A. $(0,2)$

B. $(-4,2)$

C. $(2,0)$

D. $(2,-4)$

ID: 26eb61c1

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A. $6x^2y^2\left(2x^6\right)$

B. $6x^2y^2\left(x^4\right)$

C. $6x^2y^2\left(x^6 + 2\right)$

D. $6x^2y^2\left(x^4 + 2\right)$

ID: 8f65cddc

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A. $x = \frac{7by}{11}$

B. $x = y - 77b$

C. $x = \frac{y}{77b}$

D. $x = 77by$

ID: 2926cc6d

$$(5x + 4)(2x - 5) = 0$$

Which of the following is a solution to the given equation?

A. $-\frac{5}{2}$

B. $-\frac{5}{4}$

C. $-\frac{4}{5}$

D. $-\frac{2}{5}$

ID: dd8ac009

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sara opened the account and the total amount d, in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d?

A. $d = 0.006(1 + 670)^t$

B. $d = 670(1 + 0.006)^t$

C. $d = (1 + 0.006)^t$

D. $d = (1 + 670)^t$

ID: dbe9b217

The function f is defined by $f(x) = 8x^3 + 4$. What is the value of $f(2)$?

ID: 42f19012

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

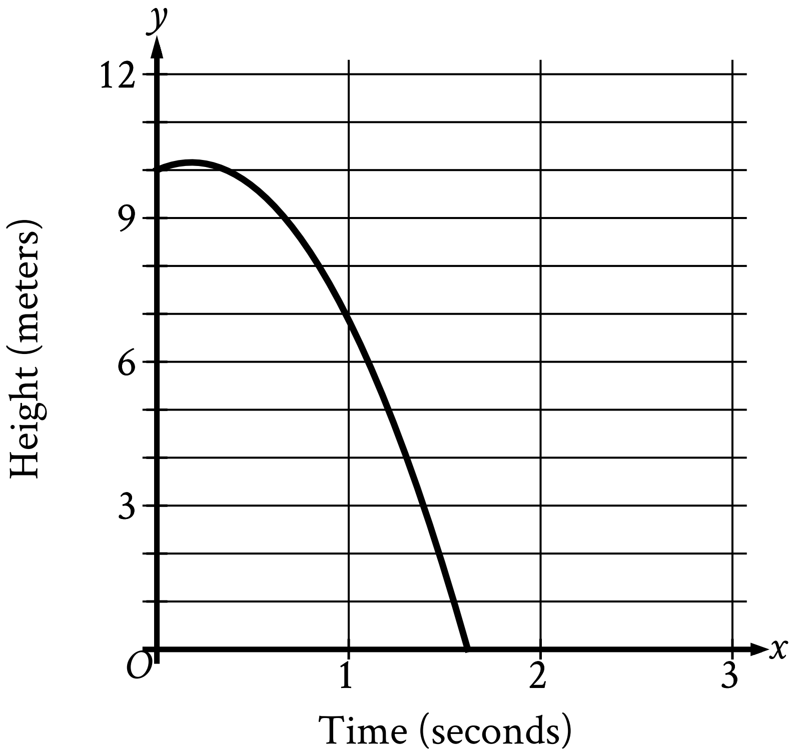
A. $\sqrt[12]{a^{132}}$

B. $\sqrt[144]{a^{132}}$

C. $\sqrt[121]{a^{132}}$

D. $\sqrt[11]{a^{132}}$

ID: 9ff88bb5



A competitive diver dives from a platform into the water. The graph shown gives the height above the water y, in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the *x*-intercept of the graph?

A. The diver reaches a maximum height above the water at 1.6 seconds.

B. The diver hits the water at 1.6 seconds.

C. The diver reaches a maximum height above the water at 0.2 seconds.

D. The diver hits the water at 0.2 seconds.

ID: 5ae4803e

$$\frac{(x+9)(x-9)}{x+9} = 7$$

What is the solution to the given equation?

A. 7

B. 9

C. 16

D. 63

ID: 281a4f3b

A certain college had 3,000 students enrolled in 2015. The college predicts that after 2015, the number of students enrolled each year will be 2% less than the number of students enrolled the year before. Which of the following functions models the relationship between the number of students enrolled, $f(x)$, and the number of years after 2015, x ?

A. $f(x)=0.02(3{,}000)^x$

B. $f(x)=0.98(3{,}000)^x$

C. $f(x)=3{,}000(0.02)^x$

D. $f(x)=3{,}000(0.98)^x$

ID: f237ccfc

The sum of $-2x^2+x+31$ and $3x^2+7x-8$ can be written in the form ax^2+bx+c, where a, b, and c are constants. What is the value of $a+b+c$?